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Are we doomed?

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AFTER THE SPIKE: THE RISKS OF GLOBAL DEPOPULATION AND THE CASE FOR PEOPLE

by Dean Spears and Michael Geruso.

Bodley Head, 307 pp., £22, July 2025, 978 1 84792 835 1

NO ONE LEFT: WHY THE WORLD NEEDS MORE CHILDREN

by Paul Morland.

Swift, 264 pp., £12.99, March 2025, 978 1 80075 412 6

THE DECLINE AND FALL OF THE HUMAN EMPIRE: WHY OUR SPECIES IS ON THE EDGE OF EXTINCTION

by Henry Gee.

Picador, 278 pp., £18.99, March 2025, 978 1 0350 3083 5

PEOPLE ARE LIVING longer than they used to. They are also having fewer children. The evidence of what this combination can do to a society is growing around the world, but some of the most striking stories come from Japan. For decades the Japanese health ministry has released an annual tally of citizens aged one hundred or over. This year the number of centenarians reached very nearly a hundred thousand. When the survey started in 1963, there were just 153. In 1981 there were a thousand; in 1998 ten thousand. Japan now produces more nappies for incontinent adults than for infants. There is a burgeoning industry for the cleaning and fumigating of apartments in which elderly Japanese citizens have died and been left undiscovered for weeks, months or years. Older people have far fewer younger people to take care of them or even to notice their non-existence. That neglect is a brute function of some simple maths. In 1950, Japan had a Total Fertility Rate (TFR) of 4, which represents the average number of children a woman might expect to have in her lifetime. Continued over five generations, that would mean a ratio of 256 great-great-grandchildren to every sixteen great-great-grandparents – in other words, each hundred-year-old might have sixteen direct descendants competing to look after them. Today Japan's TFR is approaching 1: one child per woman (or one per couple, half a child each). That pattern continued over five generations means that each solitary infant has as many as sixteen great-great-grandparents vying for his or her attention. Within a century the pyramid of human obligation has been turned on its head.

There are two different ways to describe a family tree. You can start with an individual and trace their ancestry through parents and grandparents and beyond to produce a picture of the variety of human beings it takes to make up any one of us. That's genealogy. The other version is to pick someone from an earlier generation – king or commoner – and count their direct descendants. That's heredity. But the choice between the two approaches – one spreading out backwards and one spreading out forwards – is only possible because earlier generations tended to have multiple offspring. Once that ceases to be true, family trees start to look the same from both directions. That has never happened before and it is hard to know what it will mean. But it is going to feel oppressive.

What can make depopulation difficult to fathom is the different timescales on which it operates. In one way, it has come about very quickly. In 1950 Luxembourg was the only country in the world with a TFR of less than 2, which is the magic number above which human populations grow and beneath which they start to shrink (the cut-off is actually closer to 2.1 because of infant mortality and gender imbalances, but the round figure is still a useful marker). A TFR under 2 – even if it's 1.9, Luxembourg's figure in 1950 – is called a 'below-replacement fertility rate'. Today roughly half the countries in the world have a below-replacement TFR, including almost all of the richer nations and some of the poorer ones. In 1950 many countries had a TFR of 6 or higher, including South Korea. Today just about nowhere does. In South Korea the TFR is currently 0.75, which if maintained will result in a rapidly disappearing population. Some parts of Korean society have already begun to embrace this future. The '4B' movement, which emerged on Twitter in the late 2010s, advocates for women to say 'no' four times over: no to dating; no to sex with men; no to marriage; no to childbirth.

Yet this dramatic turnaround has happened in a world in which the overall population is rising and will continue to do so for decades to come. That is because even big shifts in population trends take a long time to make a global difference. If people stop having children today but also live longer then it is only when the older generations die off that the numbers start to fall. Equally, a country's TFR can't go far below 2, which means that as long as people in some places are still having children at more than twice the replacement rate then total populations will continue to rise. In Nigeria the TFR is currently 4.5. That will almost certainly drop, in line with what is happening elsewhere. But in the meantime, Nigeria's population is expected to double over the next few decades while South Korea's halves. Before long there will be half a billion Nigerians and barely 25 million South Koreans, which adds up to a lot more people. As Dean Spears and Michael Geruso write in *After the Spike*, 'in the short run of one generation or less, the global population is a very big ship, slow to turn.' Much of what will happen is already determined, so that even dramatically declining birth rates in some places will have little overall impact. It is easy to miss the bigger picture and also to feel the futility of trying to change the direction of travel. Spears and Geruso believe that unless people start having more babies soon then we are storing up a world of trouble. But they acknowledge that even if fertility increased dramatically tomorrow the global population would be pretty much the same next year as it is today. And in a few decades there will be many more people in the world, whether we choose to have more children or not. So why bother?

In this mismatch of time frames, the depopulation problem bears a striking resemblance to the other great global challenge of our time: climate change. Many of the effects of climate change are going to play out whatever we do now. Even if we achieved net zero tomorrow the world would still get warmer for decades to come. That can make present-day action feel futile: our best collective efforts seem to have little effect on the direction of the ship. And though the problem is global, its consequences are experienced unequally. Some places suffer while others enjoy the benefits. Just as falling birth rates can be a boon for overpopulated, impoverished places, rising temperatures can be good news in the short term for colder environments. For now, a warming planet means that lives lost to heat deaths are outnumbered by lives saved as elderly people are spared the worst effects of winter. But, as with depopulation, we are getting a glimpse of what may be to come. South Korea serves as a demographic warning to the rest of the world, in the way that, say, the Maldives does for climate change. These are the places where we can see what will happen if and when the waters close over our heads.

Depopulation is at present a problem for affluent societies: in general, the richer a country the fewer children its citizens will have. Whereas a warming world will affect the poorest places on Earth first – including sub-Saharan Africa – those are also likely to be the last locations where populations will stop growing. Canada may soon be ground zero for falling birth rates – its TFR is almost as low as Japan's – but it will probably wait longer than most to experience the worst effects of climate change. In that sense, Canada is sub-Saharan Africa when it comes to depopulation and sub-Saharan Africa is Canada when it comes to climate. But these stories will also interact. In the face of declining populations and growing strains on labour forces, rich countries will become more and more dependent on immigration to maintain numbers. At the same time, as those parts of the world with growing populations become less habitable because of climate change, the impetus to move from South to North will increase. Just one of these factors on its own – either depopulation in the North or climate change in the South – could be enough to drive mass migration. Taken together they make it inevitable. Anyone who thinks the 21st century will not see the biggest global movement of peoples in history has not been paying attention.

No one can know the actual numbers involved. All future projections are guesswork. But something else that links depopulation and climate change stories is the way they both turn on the number 2. That is the figure we have often been told marks the point beyond which climate change becomes truly dangerous and difficult to control: keeping temperature rises under 2°C above the pre-industrial average is a long-term goal of international climate policy (1.5°C would be better but that ship has probably sailed). A TFR of 2 is also the cut-off for a self-sustaining population. In one case a sustainable future depends on not falling below 2 and in the other it depends on not going above it. But the climate number is essentially arbitrary: it has no significance in nature – nothing hangs on whether it's 1.9°C or 2.1°C beyond the fact that we have decided to make the figure in between the marker of our fate. But the number 2 remains the unavoidable benchmark of natural fertility. So long as reproduction requires two people to produce one child, every couple needs to have at least two children to keep the population stable. The difference between a TFR of 1.9 and one of 2.1 is like the difference between Mr Micawber's annual expenditure of 'nineteen pounds nineteen and six' and of 'twenty pounds nought and six': one leads inevitably to increase, the other diminishment.

Spears and Geruso describe the gap in demographic terms between 1.9 and 2.0 as 'profound'. But they also warn that it is easy to ascribe special significance to what is still just a number. It might be tempting to think that the unavoidable relationship between two parents and two children for sustainable human reproduction makes it a natural state of affairs. But it isn't. Societies with a TFR of 2 are very rare and not at all stable, because just about everywhere that has had that TFR has seen a rapid transition to a lower fertility rate. A country that arrives at a TFR of 2 is already on the path to something significantly lower. How fast the population shrinks is a function of how far below 2 the TFR goes. And just as a world that stabilises at a temperature rise of 5°C above the pre-industrial average will be radically different (and far worse) than one that stabilises at a rise of 2.5°C, so a world with a South Korean-like TFR of 0.75 will be very different from one with a TFR of 1.5 (the current rate in the UK). Both will lead to human extinction (something that even a temperature rise of 5°C probably wouldn't achieve on its own). But how long that will take and what it is like to experience the transition to a depopulated world will be very different in the two cases. Fixating on a particular number can distract us from what really matters, which is recognising the path that we are currently on. So Spears and Geruso suggest that 'it's time to shake any residual belief in the magic, magnetic power of two.'

But the point they want to make above all is that we have got the relationship between population and climate change all wrong. In the 1960s and 1970s, the world seemed to be confronted by the twin threats of overpopulation and ‘global cooling’: a planet drowning in people was apparently also facing a new ice age. We now know both doomsday scenarios were hopelessly misplaced. But because the population kept growing as the planet got warmer it was easy to miss what was happening. Climate change can appear to be a crisis of overproduction linked to overpopulation: more people means more mouths to feed, more industrial activity and more carbon in the atmosphere. It seemed to follow that one way to tackle the climate emergency was to have fewer children. That view is still widely held, especially among well-meaning progressives. Big families look greedy and selfish. Small families look sustainable. And maybe no children is best of all. Don’t reproduce and don’t eat meat were twin mantras for anyone who wanted to reduce their carbon footprint over a lifetime. That second piece of advice still holds. But the first was a misunderstanding. We don’t currently need fewer children. We need more.

This is where the problem of overlapping time frames really bites. Scare stories about overpopulation and global cooling were both wrong, but only one prediction went into reverse almost immediately. By the late 1980s, it was clear the planet was heating up. By contrast, the news on population has been slow to filter through. In 1968, Paul and Anne Ehrlich published *The Population Bomb*, which became an international bestseller and made Paul Ehrlich a fixture on the chat show circuit (he appeared on the *Tonight Show* with Johnny Carson more than twenty times). The Ehrlichs believed that without population controls the human species would soon overwhelm its ability to support itself. His warnings of imminent societal collapse were wild and lurid. He had been particularly affected by a visit to Delhi in 1966: ‘People eating, people washing, people sleeping. People visiting, arguing and screaming. People thrusting their hands through the taxi window, begging. People defecating and urinating. People clinging to buses. People herding animals. People, people, people, people.’ But Ehrlich had mistaken social, cultural and economic conditions for demographic ones. As one critic pointed out, the population of Delhi at the time of his visit was 2.8 million. By comparison the population of Paris in 1966 was eight million. Yet no one complained that the streets of Paris were swamped. Instead, the city was an emblem of urban sophistication. Sheer numbers weren’t the problem. It was the situation people found themselves in that mattered.

Still, the population of Delhi city today is eleven million (roughly the same as the entire Paris metropolitan area), and the number living in the larger Delhi metropolitan area closer to 35 million. Isn’t it the direction of travel that counts? Yes, which is the reason the TFR matters. India is now edging below replacement level (at 1.94) and Delhi is almost at Canada levels (at 1.4). Cities continue to grow because people move to them, not because their inhabitants are having more children. And cities are the best place to be climate-wise because energy needs can be much more efficiently met where people are gathered together. The immediate threat of climate change has little or nothing to do with population. Once upon a time it might have been possible to tell a story of breakneck economic growth and heedless exploitation of the natural environment going hand in hand with the need to feed exponentially more mouths. But that was no longer the case even when Ehrlich published his book – the rate of global population growth had started to slow in the early 1960s.

Maybe Ehrlich shouldn’t be blamed for failing to see what was happening in real time. His policy prescriptions, however, including forced sterilisation as a last resort, showed a monstrous misapprehension. China’s one-child policy stands as testament to how easy it was to

miss the underlying forces at work. China's rate of population growth was falling before the policy was introduced, continued to fall while it was in place (from 1979 until 2015), and has fallen even more rapidly since it was lifted, such that China's population will soon start to shrink. The intrusion and coercion involved in limiting the number of children a couple could have wasn't merely cruel. It was also redundant.

BUT WHY do Spears and Geruso think that having more children – and more mouths to feed – would be good for the environment? They acknowledge that if a doubling of the birth rate meant a sudden doubling of the global population it would be disastrous. In fact, there is very little that even a big shift in national birth rates could do to alter the trajectory of climate change. Decarbonisation needs to happen relatively soon – and if it doesn't it will be a failure of market economics, environmental policy and political will, not a result of demography. The next few decades are crucial. As Spears and Geruso write, 'measured at the speed of demography, 2050 might as well be tomorrow.' Nothing we do in reproductive terms will make a difference to what we achieve in environmental terms. Increasing a country's TFR from 1.5 to 2 would not affect global temperature changes. 'Fertility rates are different from day one. But almost all of the extra lives in the stabilisation path happen many decades in the future, after 2100.' The fate of the planet will be decided by what we manage – or fail to manage – in the interim.

The question of population kicks in during the interim, however, because in the absence of a move towards a stable birth rate there is a real risk that we will divert more and more of our attention and resources to the challenges of ageing societies. It would take us until the 22nd century to make the world noticeably fuller of people. But we can already see the way falling birth rates affect national politics and economics. Ageing societies vote differently, consume differently and invest differently from more balanced societies. Older people are less likely to move house. They are more likely to worry about immigration. They tend to save rather than take risks with their money. And as they become more numerous relative to other cohorts, they decide elections (even in places without elections, such as China, they have a greater influence on policymakers). A falling birth rate makes thinking about the future harder because it means a greater share of resources being directed towards the needs of people who have already lived most of their lives. This isn't a bad thing in itself. Failing to look after the needs of the elderly would be a terrible injustice. But precisely because it will be necessary it will also be constraining: it threatens to distract our attention from the other things that matter.

Spears and Geruso offer a range of arguments for having more children: young people create dynamism; larger populations are more likely to produce exceptional talent; dense urban environments are where innovation happens. And the presence of more children would help to redress a steady and subtle shift in social priorities away from young people and towards ageing populations. This has been happening for some time and is only going to accelerate. But Spears and Geruso understate the difficulty of doing something about it. One problem is that the burden of having more children will fall on people who are already holding their societies together. Parents will have to bring up extra children while looking after more elderly relatives.

Along with the TFR, another key measure for thinking about population trends is the dependency ratio: that is, the number of people of working age relative to the number who are not in the labour force (dependents), whether children or the elderly. For at least one generation, a rising TFR means a rising dependency ratio. If the number of children per typical household in Japan went back up to four tomorrow – that would mean that for the next twenty

or more years the burden of raising those children would fall on the people who are also providing for Japan's octogenarians, nonagenarians and centenarians. A rising dependency ratio is not simply something that is going to get worse before it gets better. Making it better will for quite a while be the thing that makes it worse.

There is still plenty that could be done to ease the transition. But the politics already look very hard. Resources could be directed away from pensioners and towards the needs of those raising young children. The fight in the UK over lifting the two-child benefit cap while keeping the pension 'triple lock' in place (which guarantees that pensions rise in line with wages and inflation) is a foretaste of what that contest might look like. And much more would need to change to make a significant difference to fertility rates. Massive redistribution from old to young – from healthcare for the elderly to universal childcare, from pension funds to child trust funds, from people who own their homes to people who don't – would be required. This would have to happen in societies in which elderly voters continue to outnumber younger voters, in which the immigration that is needed to counteract the short-to-medium-term effects of falling birth rates most angers those who remember a time before it was necessary, in which global competition for people and resources will put increasing pressure on living standards in those places where birth rates have fallen furthest and fastest. It may be a failure of political imagination, but I can't see it happening.

What's more, it's by no means clear what level of redistribution would be enough. Policymakers don't know what it takes to persuade people to have more children. Many schemes have been tried, from cash incentives to free childcare to better parental leave to housing support. Some of these have stopped birth rates falling as fast as they were. A few have stabilised the TFR. None has succeeded in turning a society in which the rate has fallen below replacement levels into one where it is above 2 again. That has never happened anywhere.

NOT EVERYWHERE is following the same trend. There are notable outliers which may offer clues as to what prevents depopulation from happening in the first place. In *No One Left*, the demographer Paul Morland describes two such exceptions to the global rule. One is Indonesia, which has for more than three decades existed in what Morland calls the 'Goldilocks zone' of a total fertility rate between 3 and 2: not so high as to put a strain on social and economic progress, but not so low as to become a drain on that progress. Nearby Thailand, by contrast, saw its TFR fall to 3 in the early 1980s and then keep on falling: within a decade it was below replacement levels and it is now hovering at 1.2. Before long, Thailand will have a dependency ratio of 1:1, whereas for decades to come Indonesia will have three working-age people for each person aged over 65. Morland describes Jakarta as having neither the air of an 'undernourished kindergarten' nor of a comfortable old people's home. It is full of people in their twenties 'hassling and hustling and trying to get ahead'. 'If the rest of us are going to thrive,' Morland writes, 'we have to learn something from countries like Indonesia.'

What is the lesson? What makes Indonesia different? Part of it is religion. Fertility rates in East and South-East Asia have fallen earliest and fastest in places where Buddhism prevails or has historically prevailed, including China as well as Japan and Thailand. Rates are also falling rapidly in predominantly Hindu India. Depopulation is happening more slowly in mainly Muslim societies such as Indonesia. More broadly, it is the Abrahamic religions – Christianity and Judaism as well as Islam – that tend to delay the fertility collapse. The reasons for this are contested and complex. The policy implications, however, are baffling. If politicians don't know how to persuade people to have more children, they are unlikely to know how to persuade them

to become more traditionally religious in order to do so. Indonesia may also be an outlier because of its geography. As a nation of islands – more than six thousand are inhabited – it contains many regions where Jakarta seems a world away. In remote areas, older patterns of behaviour persist, along with limited access to contraception. Indonesia's overall birth rate may have fallen more slowly simply because higher birth rates have continued in parts of the country that have been cut off from prevailing trends. At the same time, 'more islands' is not advice that is going to help anyone.

The other exception is Israel, which is unique among so-called developed nations in having a total fertility rate not simply above replacement level but pushing towards 3. What's more, Israel has achieved this despite being a highly educated society with a tech-driven economy, both of which tend to depress birth rates elsewhere (see South Korea). Again, religion seems to be an important factor, but it's a long way from being the whole story. It is true that the most religious sections of Israeli society – including Orthodox Jews – have the highest birth rates. But Judaism presents a mixed picture globally: secular Jews in the US have some of the lowest fertility rates in the country. More striking, Jews who move from societies where they share in the domestic trend not to have many children – for instance, from Russia, where the TFR is below 1.5 – start to have more children when they arrive in Israel. It's Israel, rather than Judaism, that makes the difference.

This appears to have something to do with its being a country that perceives itself to be under constant threat and is in a state of semi-permanent war. Choosing to move to such a place signals a strong interest in its survival, which means trying to ensure there are enough future generations to keep it going. Israel has other distinctive characteristics, including having one of the most generous state-sponsored IVF programmes in the world, with the highest uptake. But that is more likely to be a symptom than a cause of its pro-natal culture. The IVF programme has little effect on the overall birth rate, but it signals the value that Israel's government – and its voters – puts on population growth. Offering free IVF is not going to be a magic bullet to solve the problem of depopulation.

There is no magic bullet. Neither geographic isolation nor a sense of existential threat can be readily reproduced. Meanwhile, current policy prescriptions look like tinkering round the edges of the problem. These books want to suggest that if more of us understood the bigger picture, and recognised that we have parts of our doomsday scenario back to front, we would be less fearful of the future and keener to have children. They think too many of us have become afraid of our own shadow. But it is also impossible to avoid noticing that these are books written by men – well-meaning men, admittedly, who emphasise over and over again that they know how bad that looks. Fertility rates are measured not per person but per woman. Childbirth is done by women. Childrearing still primarily falls on women. It is women's behaviour and preferences that will need to change, even though the way men behave has a significant effect on falling birth rates (men often display greater reluctance to have children). It can seem like we are drifting into *Handmaid's Tale* territory.

Spears and Geruso, like Morland, want to make it clear that they don't condone coercion. There is no evidence that the birth rate can be forced up in the long run by taking control of women's bodies. It has been tried – for instance in Ceaușescu's Romania, which banned abortion and instituted pelvic exams for working women. Romania's TFR briefly climbed above 3 in the late 1960s but soon started falling again. Abortion went underground, many women died as a result, and in a climate of fear fertility continued its remorseless decline. Banning abortion does

nothing to increase birth rates and much to convince women that reproduction is an unsafe business.

Everything Spears and Geruso propose to increase women's willingness to have children is intended as feminist carrot not patriarchal stick: more child support, flexible career options, greater equality in domestic labour, fairer working conditions. One of the reasons the fertility rate has fallen so far in South Korea is that in a Westernised but highly patriarchal society, opting out of family life can seem like the best option. Better choices for women – rather than fewer – are the only way this is going to change.

By focusing on choice, Spears and Geruso are trying to resist depopulation determinism. There is an argument that female education means an inevitable decline in birth rates (the British historian Lucy Worsley once said she had been 'educated out of the natural reproductive function'). It is certainly true that countries which increase educational opportunities for women also see a sharp decline in the TFR; high TFRs exist only in societies with very limited female education. This is both cause and effect: educational access means more reproductive choice; more choice makes it easier to access education. But when a society's fertility falls below replacement level the story gets more complicated. There is some evidence that women with PhDs are more likely to have children than those who have undergraduate degrees. This is presumably because having an academic career, though far from ideal, has become a more child-friendly option than some other professions. A recent report identified British women who were the first generation in their family to go to university as the cohort least likely to have children. That means second and future generations might be different: to be the child of a woman who went to university shows it is at least feasible to do both. The direction of travel is not only one way.

Another pseudo-deterministic story concerns political allegiance. The problem, we're told, isn't that all humans are having fewer children, just the left-leaning ones. Liberals in the US are reproducing at a level well below replacement; conservatives, by contrast, are increasing their numbers. The top ten states for fertility are all red states (South Dakota comes first, followed by Nebraska); the bottom ten are all blue states (Vermont is last, chased by Oregon). Does this mean that liberals are eventually going to die out, to be replaced by their more fertile opponents on the right? The most extreme version of this argument suggests that religiously inspired conservative groups will inherit the earth. The Amish, for instance, who still average five or more children per family, are expanding rapidly while much of secular America contracts. In the mid-1970s there were around 50,000 Amish. There are now 400,000. At that rate of growth there will be 17 million by the end of this century and almost a billion by the end of the next. Except it doesn't work like that. Small communities can grow relatively quickly, but after a certain point scale kicks in and birth rates fall. What's more, as communities grow, more children leave them, and when they do they start to reproduce like Americans, not like Amish.

All sorts of reasons have been put forward for conservatives having more children than liberals (it is also possible that having children makes people conservative). But, over time, as the cost of caring for children in ageing societies starts to rise, this asymmetry could change. Having children in a depopulating world is likely to make parents feel angry with a system that fails to respect their choices. In 2017, when Jeremy Corbyn came close to winning power in Britain, his surprising level of electoral support was originally described as a 'youthquake'. But that turned out to be a misnomer, since there weren't enough young voters to make a difference. It was the parents of young children who turned against the Tories, having had enough of a party that

primarily served the interests of the elderly. Politics is no more deterministic than depopulation. At present, the pro-natal argument has been hijacked by figures on the alt-right such as Elon Musk (fourteen children and counting), which can make it seem as if harping on about the birth rate is a right-wing thing to do. But that's just squeamishness on the part of the left. The fact that the argument has been hijacked suggests at the very least that there is an argument to be had.

For now, however, the direction of travel looks set. One puzzle about depopulation is that the citizens of wealthy societies, where it is happening fastest, often say the reason they don't have children – or delay it for as long as possible – is because they can't afford it. How can it be that getting richer also makes people feel they are too poor to reproduce? The answer, as Spears and Geruso point out, is opportunity costs. As societies become more affluent, there is a far greater premium placed on space and time, especially for people who lead busy, metropolitan lives. The price of many things – painkillers, solar panels and televisions – goes down over time. By contrast, 'products where the main ingredient is people ... do not get cheaper over time in the same way. That's because relative to other ingredients, people's time is expensive.'

Parenting costs time – lots of it. And that's the thing we think we can least afford to lose. More choice in the way we live means more reasons to worry that having children will limit our choices. And greater expectations about what a decent life for our children would be means that having more children is a greater compromise. Take something as seemingly banal as the question of whether a child has a room of his or her own. If your children share bedrooms, then an extra child is not necessarily a big extra cost: you won't have to move house or build an extension. But if each child means another bedroom, you are going to think twice. The average American child in 2020 lived in a house with 5.6 rooms; the average home in 1960 had 4.4 rooms and held more children. More space doesn't mean more space for kids. It means a bigger sacrifice each time another one comes along.

Spears and Geruso put it this way: 'A better world, with better options, makes parenting worse by comparison.' It's a bleak finding for anyone who wants to suggest this is something we can change. To produce a positive effect, the improvements will have to be highly targeted and will need to focus on buying individuals a lot more time. In an era when time is an ever scarcer resource, that seems implausible. This is not a new story caused by new technology. It has been true for as long as birth rates have been measured, going back many centuries. The massive expansion in human numbers since the beginning of the 19th century – when we escaped from the Malthusian trap – was not achieved by people having more children. Long before that they had already started to have fewer.

What actually caused the growth in human population was a dramatic improvement in survival rates, first of all among children and more recently among the elderly. As infant mortality dropped from one in three to one in three hundred, which is its level in many places, overall numbers grew rapidly even as the average household was shrinking from eight to seven to six to five to four members. The first trend was simply happening much faster than the second, though the second started earlier. The infant mortality rate can't go much lower in rich parts of the world, which means the decline in the birth rate is currently happening far faster than the rise in the survival rate. At the same time, gains at the other end of the scale are increasingly marginal. Rising life expectancy tails off once most people start living into their late seventies and beyond. Reaching a hundred is much more common than it used to be, but it is still rare. Reaching 120 has only happened to one person in history. Even if we started living to 150 or

more, populations would continue to shrink so long as we continued to reproduce at below replacement levels. Long life plus few children equals human extinction in the end.

This suggests that the depopulation scenario was set in motion a long time ago. The problem of the opportunity costs of parenthood may simply be a function of modernity: as soon as humans had choices that they cashed out in terms of personal utility, reproduction became less appealing. That was already happening in the 16th and 17th centuries, when records show falling birth rates in the places where the modern world was beginning to reveal itself. The link between modernity and declining fertility could be explained by the rise of individualism or liberalism or market economics. It doesn't much matter – this is who we are. The process was slow to start with but in the 20th century it began to pick up speed. It is possible that the end of modernity could reverse it – a return to an age of superstition and magic, or alternatively a post-human future in which reproduction is freed from pregnancy could conceivably see the creation of new humans again outpace the disappearance of old ones. There are glimpses of both prospects at present but neither seems imminent, and either would cause far more disruption to our ways of living than most of us would be willing to countenance. The modern world still has a lot of life left in it, which means that human life on Earth is likely to continue its disappearing act.

DOES THAT MEAN we are doomed to die out? Henry Gee thinks so, if only because all species die out in the end. The question is simply how long it takes. The Copernican principle, which implies that the place in which you happen to find yourself in time is highly unlikely to be a special moment in history (just as it was always improbable that the Earth was at the centre of the cosmos just because it's where we live), is one way of trying to estimate how near we are to the end. *Homo sapiens* is approximately 315,000 years old. If it's heavily odds against that we are either at the very beginning (first 2.5 per cent) or at the very end (last 2.5 per cent) of the human story then our version of humanity has somewhere between eight thousand and twelve million years left to go. That's not much of a guide, but as Gee points out, we do find ourselves at the only point in the history of the species when the rate of population growth has dramatically slowed and is about to go into reverse. So maybe there is a good reason to think this is an important – and ominous – moment. Seen from that perspective, even eight thousand years looks hopeful.

There are other clues. One marker of impending demise occurs when a species becomes dominant in its environment. Becoming the top dog is a bad sign because it means that your habitat suits you too well – changes in that habitat will be commensurately worse for you than for species whose existence is more of a struggle. Similarly, ending up in an unchallenged position tends to mean that you have eliminated the nearest competition, after which interbreeding with your rivals ceases and you are left to your own devices. From that moment on your days as a species are numbered. Gee compares the turning point in the human story with the starting point of Gibbon's *Decline and Fall of the Roman Empire* during the reign of the Emperor Trajan, when Rome's sway over the known world was at its greatest extent. Once the Romans had reached their apogee, the only way was down.

For the species as a whole, it happened 40,000 years ago. That was when *Homo sapiens* became the last species from the genus *Homo* on Earth, having finally eliminated the other types (all the many versions of the original *Homo erectus*, including *Homo floresiensis*, *Homo habilis*, *Homo heidelbergensis*, *Homo naledi*, *Homo rhodesiensis*, along with the Neanderthals and countless other human variants that we know nothing about because they left no trace) and spread to every

corner of the habitable world. We were the kings of our domain. We were also doomed from then on – no one to interbreed with, no one to push us back, no one to stop us doing what we wanted with the natural world, nothing to prevent us from making ourselves more and more vulnerable to the environment we were creating. Most of the things we did subsequently we called progress. We were kidding ourselves. We were merely staving off the inevitable.

The reason *Homo sapiens* came out on top was our ability to outsmart other species (hence the vainglorious name we have given ourselves). But those powerful brains exist on top of spindly and clumsy bodies, which we have made increasingly feeble. Because *Homo sapiens* went through a series of near extinction level events before eventually triumphing – periodically reduced to a few thousand surviving members clinging on in a handful of African enclaves – we have very little genetic variation. The human family tree spreads out from a few individuals to encompass all of us. That means we are all vulnerable to the same diseases, which can move rapidly through the entire global population – something that is more likely to happen today, given the ease with which we move around the globe.

But the real enfeeblement began ten thousand years ago with the transition from hunter-gatherer to agricultural societies. Until then, the growth in human population had been limited by the practice of waiting until a child was weaned before having the next one: transporting too many babies around was a needless burden. As a result, overall human numbers topped out at around ten million until we started staying in place and farming the land. At that point two things happened. First, it became possible to have larger families. Second, it became necessary because our new way of life was a deathtrap. Settled communities were breeding grounds for disease, not helped by regular crop failures and the absence of sanitation. Worse, the domestication of many animals allowed their viruses to spread to humans. As communities got bigger, the problems got worse. Cities were killing zones even without the presence of farm animals. Gathering so many of a species so vulnerable to infection in such cramped conditions made survival a lottery. The only way the human project could be kept afloat was by having enough offspring to outrun the grim reaper. It was a miserable business for everyone, but particularly for women and for children. Human progress was a vale of tears.

Only very recently have we worked out how to do it better. We can immunise ourselves against diseases – though we remain as vulnerable to infection as ever – and we can sanitise our dwelling places. We have better access than ever to food and shelter and education. And as soon as we worked out how to do all that we stopped having children again. There is no moment in the human story when safe and prosperous living conditions have coexisted with above replacement-level fertility. When it stopped being a battle for survival it immediately became another battle, one that we are losing because we have chosen it as the less bad option: better the comfy old people's home than the hungry kindergarten. Gee believes the only way our decline could be reversed is to recreate the conditions that existed more than forty thousand years ago. We need a greater range of human variants existing under a wider variety of living conditions. That, he thinks, could happen in space, as human colonies diverge from one another to produce new versions of the species. It will have to be an almighty struggle and there will, he admits, 'be untold tragedy and heartache along the way'. All in all, it's not much of a prospectus.

In the meantime, some people are placing their hopes in AI, not merely to solve our problems but to supplement our ever dwindling numbers with human-like entities that can do the heavy lifting. If we don't want lots more immigrants – and in any case there soon won't be enough

immigrants to go around – then it will have to be the robots. Again, it's not a great lookout. But we may not get that far anyway. Though Gee doesn't discuss it at length, there is another obvious reason for thinking we might be at the end of the human story. Only in the last hundred years did we become the first species to construct the means of its own annihilation. Depopulation is starting to happen in a world in which the number of nuclear weapons is once again increasing. The old men who run the show hold the future of a dying species in their trembling hands. I'm not sure we're going to make it to the spaceships.

Letters

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David Runciman, writing about population decline, assumes that the population growth required to prevent human extinction (the ‘replacement rate’) is best described by the Malthusian formula ‘one couple = two children’ (LRB, 20 November). He may have in mind white, European, monogamous couple-based families, but in general human reproduction simultaneously requires much more and much less than a couple. On the one hand, conceiving and delivering a baby can involve more than two people (surrogate parents, sperm donors, online fertility supplement markets, the vast range of formal and informal institutions from fertility clinics to birthing and maternity wards, doulas and adoption regulators); on the other, it needn’t require a couple at all. More important, to think of reproduction rates as decided by the ratio of children to presumptive progenitors is misleading: humans are uniquely ill-equipped to survive without extensive networks of support at least until late infancy (and, as many parents of millennials and Gen-Zs will know, possibly much longer). In this sense, the replacement rate would be better defined as something like ‘approximately two people alive on Earth for every two people that were alive one generation prior’.

From this perspective, the picture begins to look very different. Not only is there no discernible threat of extinction of the human species as such, but the decline of population looks set to be governed by factors other than the rate of reproduction: wars and other forms of violence, drought and flooding caused by climate change, the collapse of food systems and so on. The picture looks more different still when we realise that when the term ‘replacement rate’ is used, it isn’t even to mean ‘for every two humans alive on the planet now, two humans alive a generation later’, but instead something like ‘for every two humans likely to rely on pensions or public health in x years’ time, two humans entering the labour force (and paying contributions) now’. If it takes a village to raise a child (or two), it takes a lot more to raise a taxpayer: usually, a nation-state or other territorially bounded political entity with a system of public collection, distribution and enforcement.

This is where the true parochialism of the ‘two for two’ formula becomes apparent. There are societies that do not value people only or primarily for their ability to engage in productive (waged) labour or provide reproductive (usually unpaid, or at least worse paid) care labour. These societies do not treat elderly members as a burden on the tax, public health and care systems; many Indigenous cultures of the Americas are examples of this, as is China. It is a feature of this increasingly violent phase of capitalism that elders are treated as a burden; there are modes of production, redistribution and exchange that do not require the increased extraction of labour power to justify maintaining or improving living standards for those outside the confines of state-sanctioned social reproduction.

One thing that Runciman infers correctly from the data is that, on the whole, people tend to reproduce less (if at all) once there is no pressure to do so. What the books he is reviewing seem to conclude is that people (and women in particular) should be persuaded otherwise. Even if we discount the horrid ways in which this is already happening – in the US, for example, not only in the states that have made abortion illegal, but also through the reduction of access to free, safe and universally available abortions – how about concluding that having children isn’t so great after all?

There would be nothing wrong with there being fewer humans, especially in the global North, whose population also tends to weigh heaviest on the environment. As degrowth theorists have argued for years, we need to produce and consume less. Unless we rapidly find a way to do that, we are likely to see massive population shrinkage anyway, but not of the voluntary sort. This, as we know, is likely to have the first and greatest impact on those least responsible for climate overshoot.

Jana Bacevic
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David Runciman, writing about the alarming inverse proportionality between a society's wealth and its birth rates, is correct that affluent societies, by placing such a high premium on human time, make childrearing a comparatively unattractive proposition (LRB, 20 November 2025). He makes only passing reference, however, to what is, as I see it, the most promising long-term fix for our demographic woes: changing our understanding of kinship.

I'm a thirty-something queer woman. Hardly anyone in my demographic lives in a straightforward single-couple household. We place less and less stock in monogamy. I have multiple romantic partners, and have various relationships – at different degrees of intimacy and physical remove – with *their* romantic partners. The question of 'romantic' partnership is itself coming under scrutiny: must we be in love to cohabit or share a bank account? What if we're just really good at working together? This sort of thinking has led to increasingly interconnected social webs, which can result in surprising economic efficiencies. For those of us now considering childrearing, these new kinship networks are starting to look like the only villages capable of raising our children.

I am aware that I'm essentially microwaving Maurice Godelier's argument in *The Metamorphoses of Kinship*: single-family households are a historical anomaly, and other arrangements are likely to have better economics. It's a mistake to hold the cost of caring for our young and elderly as essentially constant.

Violeta Cortés
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As a biologist who accepts Darwin's theory of evolution by natural selection, I was surprised by David Runciman's analysis of population decline (LRB, 20 November 2025). A species does not become extinct because it produces too few offspring. Extinction happens because the environment changes. New species arise owing to genetic mutation in offspring, some of which – by chance – can survive in the new conditions. The process happens at a glacial speed, reflecting the rate at which the environment used to change. If we are doomed, it is because our actions are causing that rate of change to increase. Look at it this way: mastodons weren't doomed because they had 'too few offspring'. The climate warmed; human hunters helped them on their way. As humans die out because it's just too hot, wet or windy for us, non-global species will heave a sigh of relief at our absence as they migrate to places more amenable to them. Mastodons evolved into elephants; perhaps humans will evolve into a less self-centred and destructive species.

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